

Raw phi1 voltage calibration

RUN

200242_raw-phi1_fine-brackets-3s

Status: interrupted; 381 individual PAX readings

MEASUREMENT

0.0 to 0.8 RP command V, step 0.02 V; 1 forward/reverse cycles.

Settling: 3.0 s; 5 consecutive readings per held level.

EXPLICIT VOLTAGE LEVELS

0,0.1,0.2,0.3,0.32,0.33,0.335,0.34,0.345,0.35,0.355,0.36,0.365,0.37,0.375,0.38,0.385,0.39,0.395,0.4,0.405,0.41,0.415,0.42,0.425,0.43,0.435,0.44,0.445,0.45,0.455,0.46,0.465,0.47,0.475,0.48,0.5,0.6,0.7,0.8

PHYSICAL SETUP

Physical position awaiting confirmation; no assumed port

IN1/IN2 physical mapping awaiting confirmation

WHAT IS RECORDED

Requested RP voltage, ASG register readback, full FPGA OUT1 scope trace, explicitly selected IN1/IN2 scope traces, raw PAX records and acquisition times.

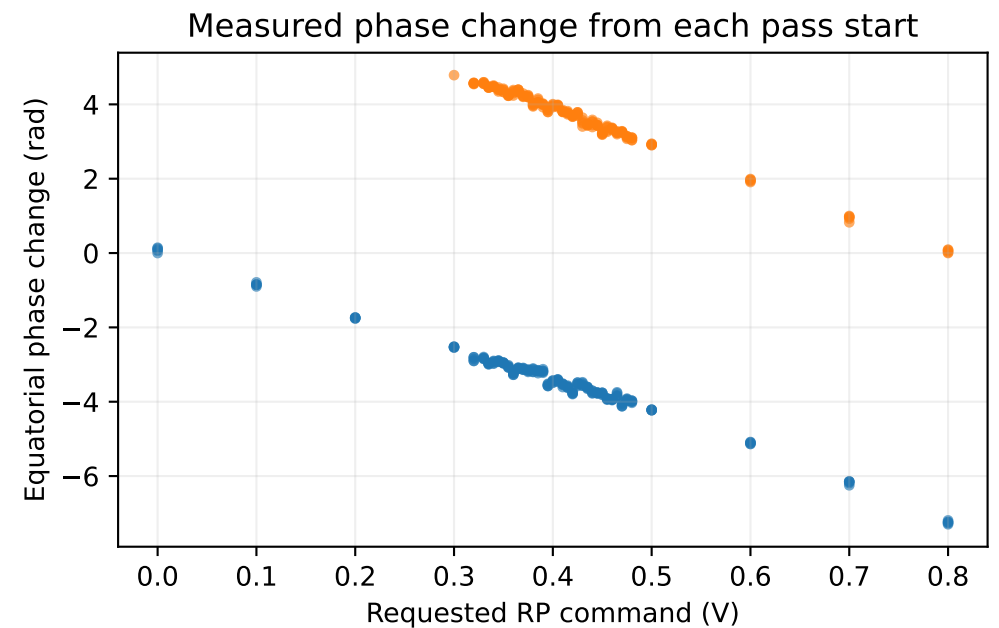
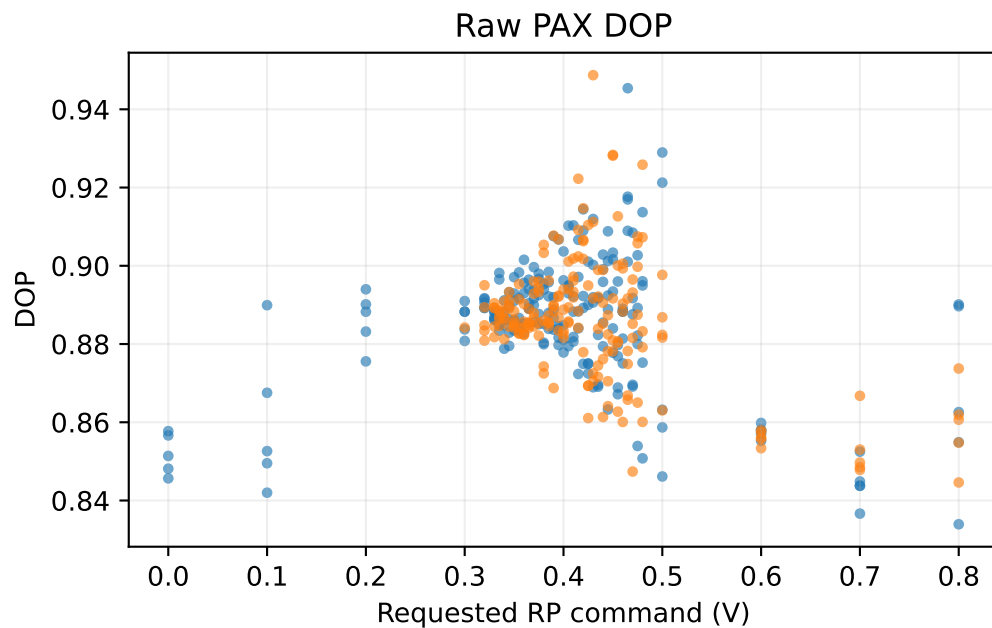
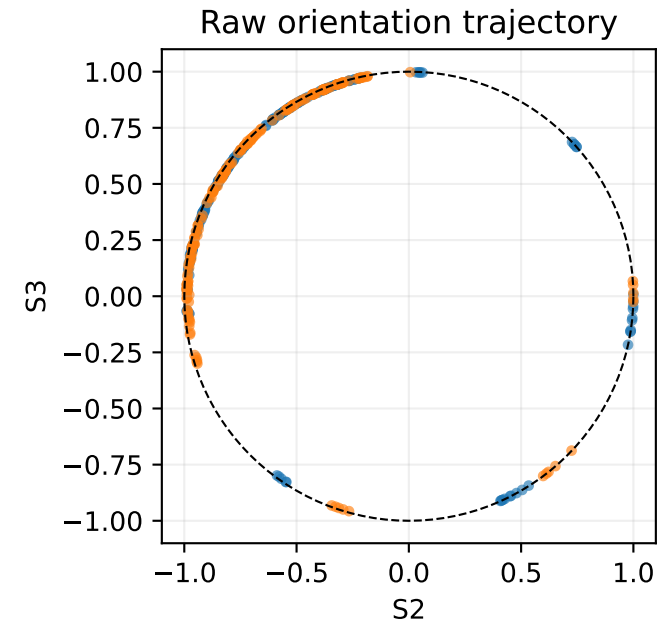
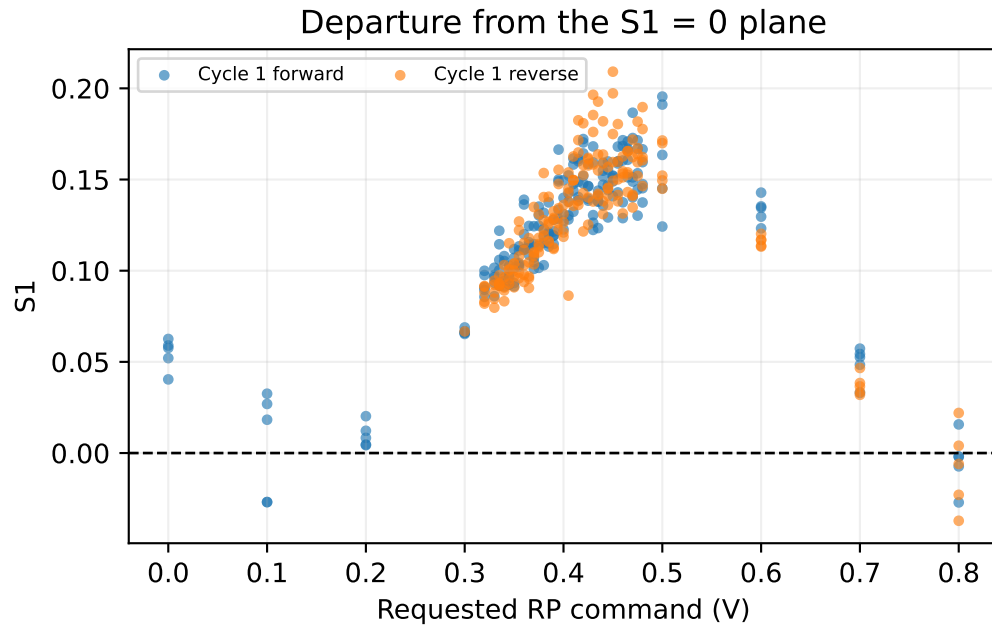
WHAT IS NOT ESTABLISHED

Internal OUT1 data is the signal before the DAC. It is not an independent BNC or piezo-terminal voltage measurement. No historical driver gain is applied.

ANALYSIS POLICY

No curve fits, interpolated V_{π} , drift subtraction, or optimized coordinate rotation. Phase is $\text{atan2}(S3, -S2)$; unwrapping only adds 2π at coordinate wraps. Step medians are labelled summaries; all individual measurements remain in data.csv and pax_raw.jsonl.

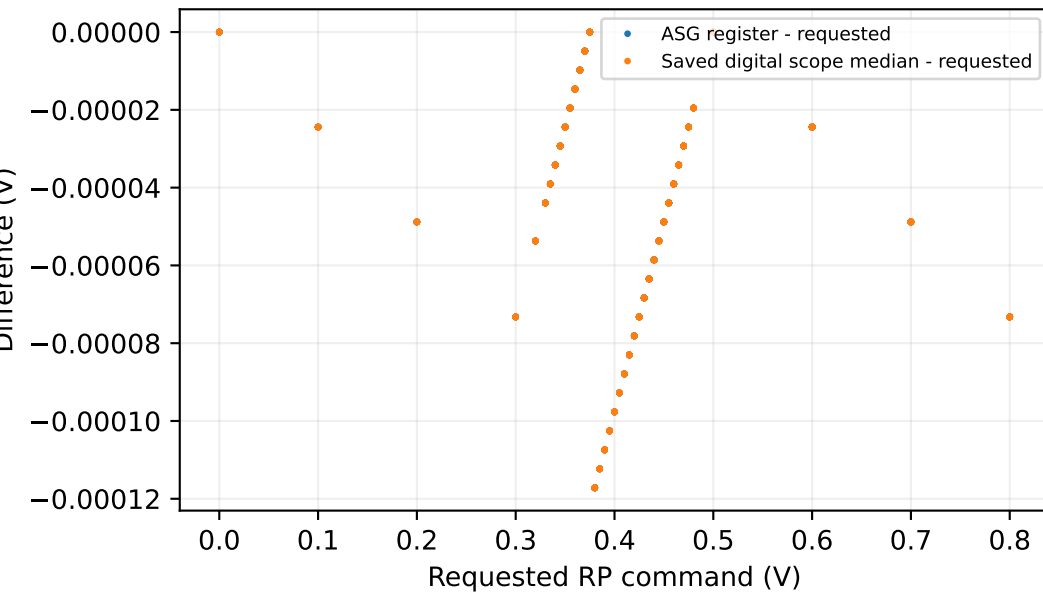
Individual PAX readings at held voltages



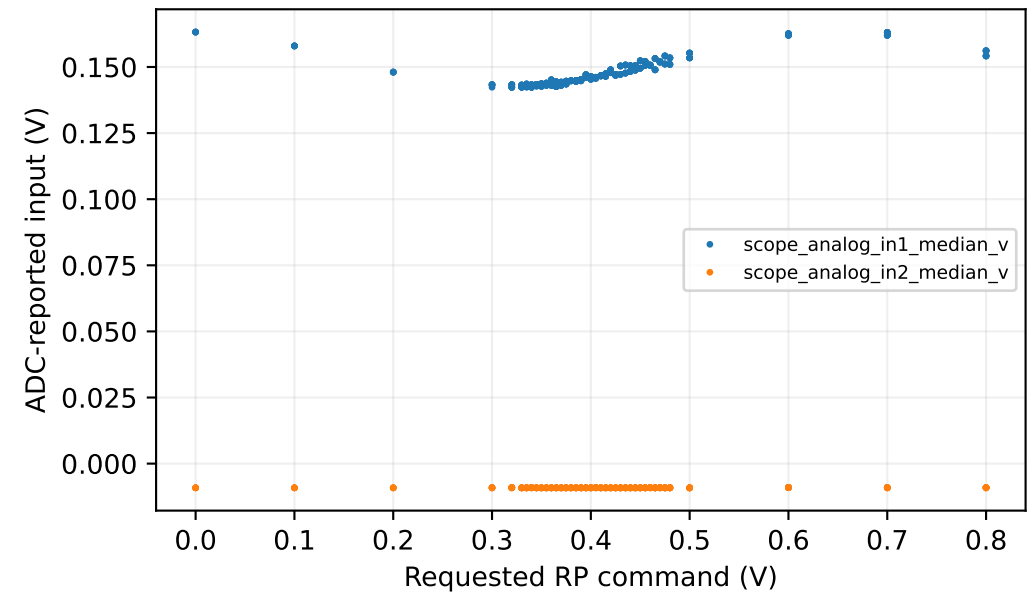
Orientation Stokes are unit-normalized from PAX angles; DOP is shown separately. No fitted or corrected trajectories.

Readbacks and direct return-path comparisons

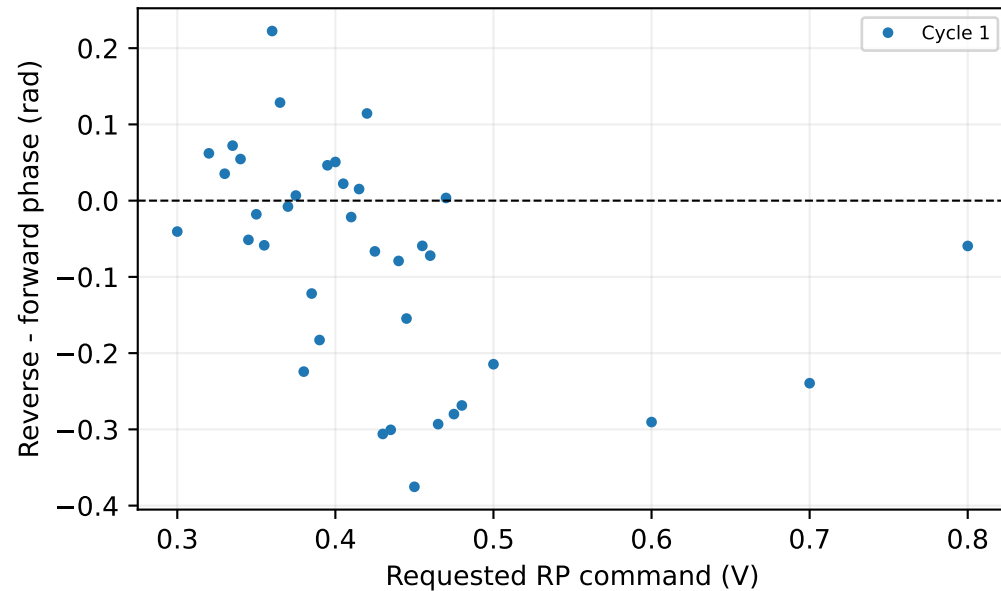
Software command versus internal readbacks



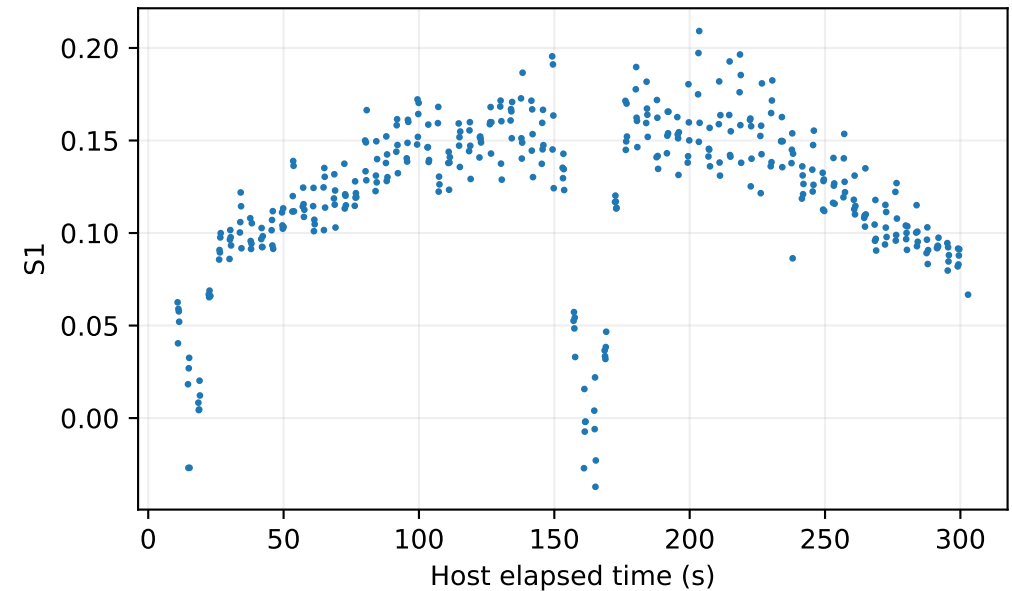
Explicitly routed analog input scope records



Same-voltage return difference: drift + hysteresis



Time record — no drift subtraction



Observed pi-crossing brackets

CYCLE 1 FORWARD — REFERENCE 0.000 V

Δ RP command 0.3500–0.3550 V; observed excursion 3.045–3.150 rad

Bracket including all observed reference/endpoint sample spreads: 0.3500–0.3600 V.

CYCLE 1 REVERSE — REFERENCE 0.800 V

Δ RP command 0.3250–0.3300 V; observed excursion 3.077–3.209 rad

Bracket including all observed reference/endpoint sample spreads: 0.3200–0.3300 V.

INTERPRETATION

Each interval brackets π between adjacent measured step medians; no interpolation is applied. Drift, hysteresis and finite step size remain in the result. These are RP command-voltage brackets for projected equatorial phase, not calibrated terminal volts or an independent measurement of actuator retardance. Departure from $S1=0$ limits the ideal-model interpretation.

Active acquisition parameters

PROBE

False

START

0.0

STOP

0.8

STEP

0.02

REPEATS

1

LEVELS

0,0.1,0.2,0.3,0.32,0.33,0.335,0.34,0.345,0.35,0.355,0.36,0.365,0.37,0.375,0.38,0.385,0.39,0.395,0.4,0.405,0.41,0.415,0.42,0.425,0.43,0.435,0.44,0.445,0.45,0.455,0.46,0.465,0.47,0.475,0.48,0.5,0.6,0.7,0.8

SETTLE

3.0

SAMPLES

5

HOLD_VOLTAGE

None

HOLD_POINTS

20

LABEL

Active acquisition parameters

fine-brackets-3s

PAX_LOCATION

Physical position awaiting confirmation; no assumed port

INPUT_CONNECTIONS

IN1/IN2 physical mapping awaiting confirmation

PAX SETTINGS

Wavelength 830.0 nm; daemon mode 5; configured rotation 60.0 Hz

CLEANUP EVIDENCE

[0.0, 0.0]